

Home Credit AI - Project Description

Project Overview

Home Credit AI is an end-to-end credit risk assessment platform. A LightGBM machine learning model predicts the probability of loan default, while SHAP Explainable AI explains every prediction. The system supports both customers and bank staff.

Portal 1: Customer Portal

Customers can: Check their loan eligibility by entering personal and financial information. View default probability, risk level, recommendation, and SHAP explanation. If satisfied, apply for the loan. The system generates a Loan ID and stores the application with Pending status.

Portal 2: Bank Portal

Bank staff log in securely and search applications using Loan ID or Customer ID. They can view customer details, ML prediction, SHAP explanation, and risk score before approving or rejecting the loan. Staff can also add remarks, after which the application status becomes Approved or Rejected.

Customer Status Check

Existing customers can check their application using Loan ID or Customer ID. The portal displays the application status, bank remarks, loan amount, and application date.

New User Eligibility Check

If a person has never applied before and does not have a Customer ID or Loan ID, they can still check their eligibility by entering their information. The ML model predicts the default probability and provides a recommendation. If they decide to continue, they can submit the application and receive a new Loan ID.

Input Features

Personal: Name, Age, Gender, Marital Status, Education, Employment Type, Years Employed, Family Members, Children.

Financial: Annual Income, Loan Amount, Loan Annuity, Goods Price.

Assets: Own House, Own Car.

Housing Type and other important details required by the model. Remaining engineered features are generated automatically.

Model Details

Model: LightGBM

Dataset: 307,511 customer records

Features: 193

Validation Accuracy: 73.83%

ROC-AUC: 0.7871

Recall: 69.04%

Explainability: SHAP

Workflow

New User/Existing Customer → Eligibility Check → ML Prediction → Apply for Loan → Loan ID Generated → Application Stored → Bank Reviews → Approve/Reject → Customer Checks Final Status

Technology Stack

Frontend: React + Tailwind CSS Backend: Node.js/Express ML API: FastAPI Database: PostgreSQL ML: Python, LightGBM, Scikit-learn, SHAP, Pandas, NumPy

Future Scope

Modern UI, authentication, analytics dashboard, PDF reports, customer history, deployment, and production-ready integration.